一個驚人的證明：$1+1=2$

求: $1+1=?$

$$= \sin\frac{\pi}{2} + \cos 0$$

$$= \sin x\Big|_0^{\frac{\pi}{2}} + \cos x\Big|_{\frac{\pi}{2}}^{0}$$

$$= \int_0^{\frac{\pi}{2}} \cos x\,dx + \int_{\frac{\pi}{2}}^{0} -\sin x\,dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x + \cos x)\,dx$$

$$= \int_0^{\frac{\pi}{2}} \sqrt{2}\sin(x+\frac{\pi}{4})\,dx$$

$$= -\sqrt{2}\cos(x+\frac{\pi}{4})\Big|_0^{\frac{\pi}{2}}$$

$$= -\sqrt{2}\left(\cos\left(\frac{\pi}{2}+\frac{\pi}{4}\right) - \cos\frac{\pi}{4}\right)$$

$$= -\sqrt{2}\left(-\sin\frac{\pi}{4} - \cos\frac{\pi}{4}\right)$$

$$= \sqrt{2}\times\sqrt{2}\sin\left(\frac{\pi}{4}+\frac{\pi}{4}\right)$$

$$= 2\sin\left(\frac{\pi}{4}(1+1)\right)$$

Let $1+1=x$

$$x = 2\sin\left(\frac{\pi}{4}x\right)$$

圖像法解方程 (見第二頁)

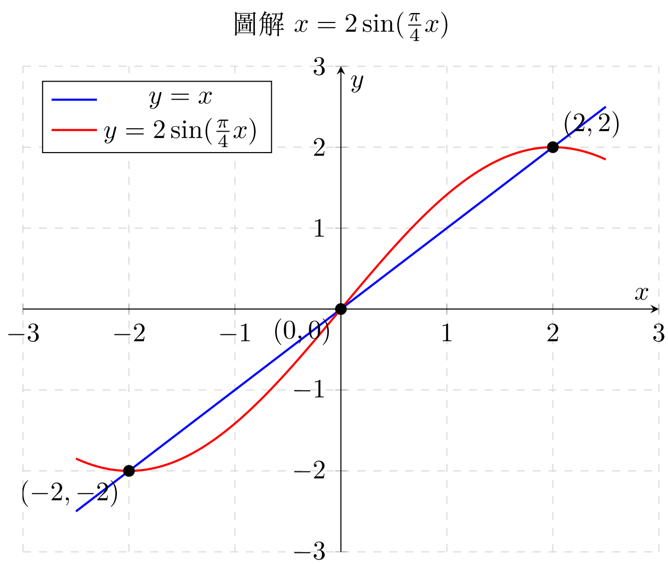

圖 1: 利用圖像法確認實數解

$$
\begin{aligned}
&x = -2, 0, 2 \\
&1 + 1 = -2, 0, 2 \\
&1 + 1 \geq 2\sqrt{1 \times 1} = 2 \\
&1 + 1 = 2_{\#}
\end{aligned}
$$